

Why Cell Age?

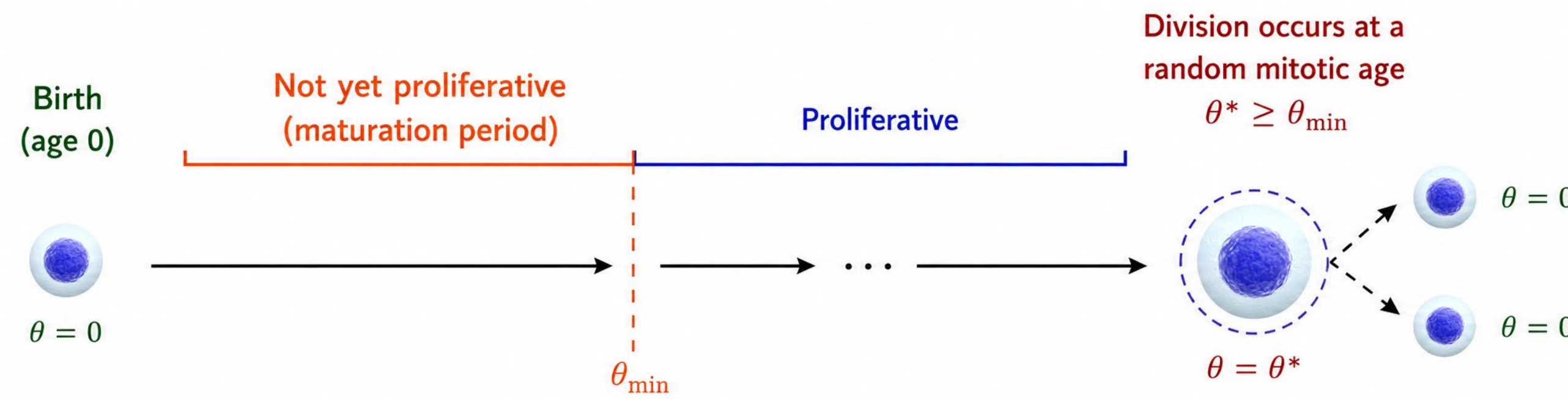
- Classical mechanical tumor models track spatial density, but not where cells are in their life cycle.
- Cell age affects proliferation and death, linking age structure to tumor growth.
- Age-structured spatial model allows us to ask how age-dependent cellular behavior shapes tumor expansion and treatment response.

Model Structure

Cell density : $n = n(x, \theta, t)$, x : space, θ : cell age, t : time.

$$(0.1) \quad \begin{cases} \partial_t n + \underbrace{\partial_{\theta} n}_{\text{age progression}} - \underbrace{\operatorname{div}_x(n \nabla p)}_{\text{pressure-driven movement}} = - \underbrace{\nu(\theta, p)n}_{\text{age-dependent division}} - \underbrace{\mu(\theta)n}_{\text{age-dependent death}}, & x \in \mathbb{R}^d, \theta > 0, t > 0, \\ n(x, 0, t) = 2 \underbrace{\int_0^{\infty} \nu(\theta, p)n(x, \theta, t) d\theta}_{\text{cell division}}, & x \in \mathbb{R}^d, t > 0, \\ n(x, \theta, 0) = n_{\text{in}}(x, \theta), & x \in \mathbb{R}^d, \theta > 0. \end{cases}$$

where the volume density is defined as $\rho(x) = \int_0^{\infty} V(\theta)n(x, \theta, t) d\theta$ and the pressure is defined as $p = \frac{m}{m-1}\rho(x, t)^{m-1}$.



Existence of a Weak Solution

Theorem 0.1. For all $m > 2$ and for any initial condition $n_{\text{in}}(x, \theta)$ such that

$$n_{\text{in}} \in L^1 \cap L \log L(\mathbb{R}^d \times (0, \infty)), \quad (|x|^2 + \theta)n_{\text{in}} \in L^1(\mathbb{R}^d \times (0, \infty)), \quad \rho_{\text{in}} \in L^{\infty}(\mathbb{R}^d)$$

there exists a weak solution $n(x, \theta, t)$ of (0.1). The function $n(x, \theta, t)$ is such that

$$n \in L^{\infty}(0, T; L^1 \cap L \log L(\mathbb{R}^d \times (0, \infty))), \quad (|x|^2 + \theta)n \in L^{\infty}(0, T; L^1(\mathbb{R}^d \times (0, \infty)))$$

while the density $\rho(x, t)$ and pressure $p(x, t) = \frac{m}{m-1}\rho(x, t)^{m-1}$ are such that

$$\rho \in L^{\infty}(0, T; L^1 \cap L^{\infty}(\mathbb{R}^d)), \quad \nabla p \in L^2(0, T; L^2(\mathbb{R}^d))$$

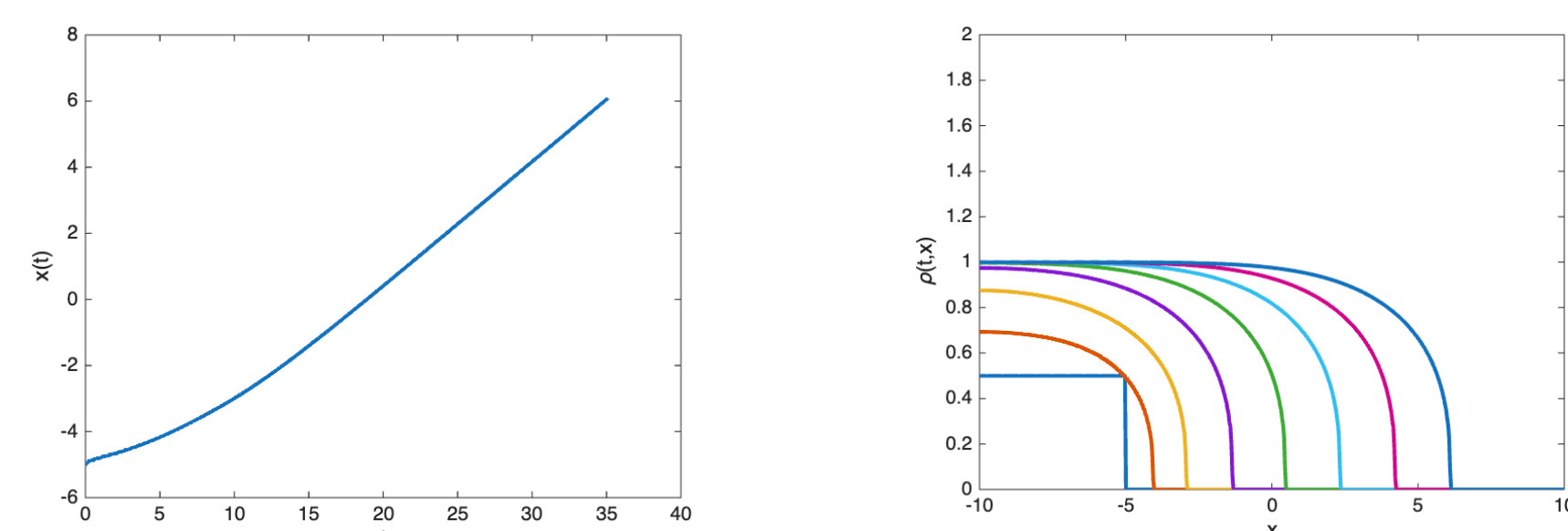
and the equation (0.1) is satisfied in the following distributional sense:

$$\begin{aligned} & \int_0^T \int_{\mathbb{R}^d} \int_0^{\infty} n(x, \theta, t) \left[-\partial_t \psi(x, \theta, t) - \partial_{\theta} \psi(x, \theta, t) + \nabla_x p(x, t) \cdot \nabla_x \psi(x, \theta, t) + \nu(\theta, p(x, t))\psi(x, \theta, t) \right] d\theta dx dt \\ & = \int_{\mathbb{R}^d} \int_0^{\infty} n_{\text{in}}(x, \theta)\psi(x, \theta, 0) d\theta dx + \int_0^T \int_{\mathbb{R}^d} \psi(x, 0, t) 2 \int_0^{\infty} \nu(\theta, p(x, t))n(x, \theta, t) d\theta dx dt \end{aligned}$$

for all $\psi \in \mathcal{D}(\mathbb{R}^d \times [0, \infty) \times [0, T))$. Furthermore, the density $\rho(x, t)$ solves

$$\partial_t \rho - \operatorname{div}(\rho \nabla p) = \int_0^{\infty} V'(\theta)n(x, \theta, t) d\theta + \int_0^{\infty} \nu(\theta, p)(2V(0) - V(\theta))n(x, \theta, t) d\theta.$$

Finite Speed of Propagation

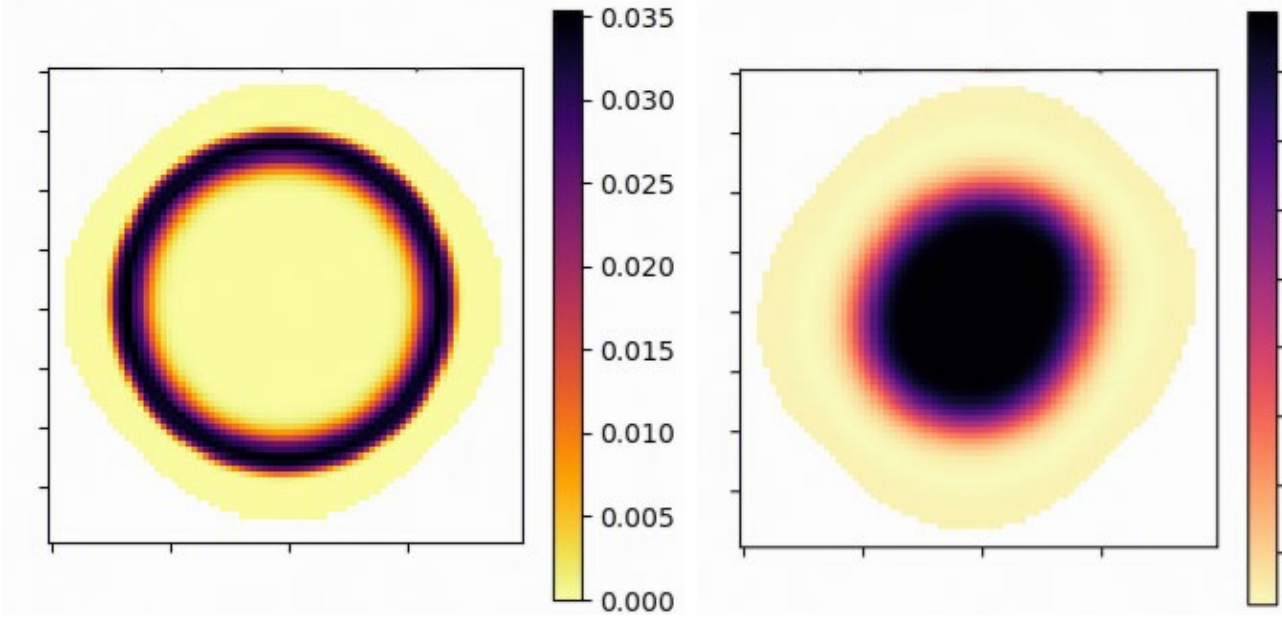


Proposition 0.2 (Finite speed of propagation of the support). Under the assumptions of Theorem 0.1, assume further that there exists R_0 such that $n_{\text{in}}(x, \theta) = 0$ for all $|x| \geq R_0$ and $\theta \geq 0$. Then

$$n(x, \theta, t) = 0 \quad \text{for all } |x| \geq \bar{R} + e^{(m-1)Ct} \text{ and for all } \theta \geq 0$$

for some $\bar{R}$ depending on R_0 and $\|\rho_{\text{in}}\|_{L^{\infty}}$.

Spatial Age Structure



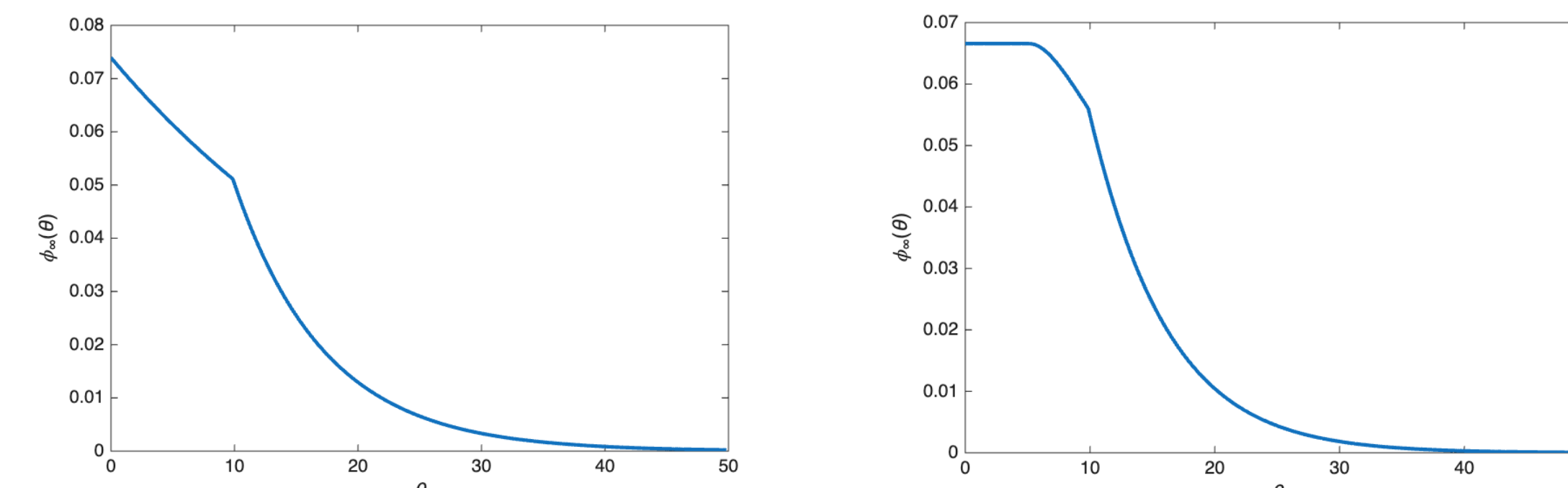
Spatial distribution of newborn cells and average cell age.

- The proliferating rim lies near, but slightly inward from, the outermost tumor boundary, whereas the oldest cells accumulate in the tumor core.

Long-Time Limiting Distribution in Age

The behavior of $n(x, \theta, t)$ for large t numerically satisfies

$$n(x, \theta, t) = \bar{n}(x - ct, \theta) \sim \phi(x + c(\theta - t)) \text{ as } t \rightarrow \infty$$

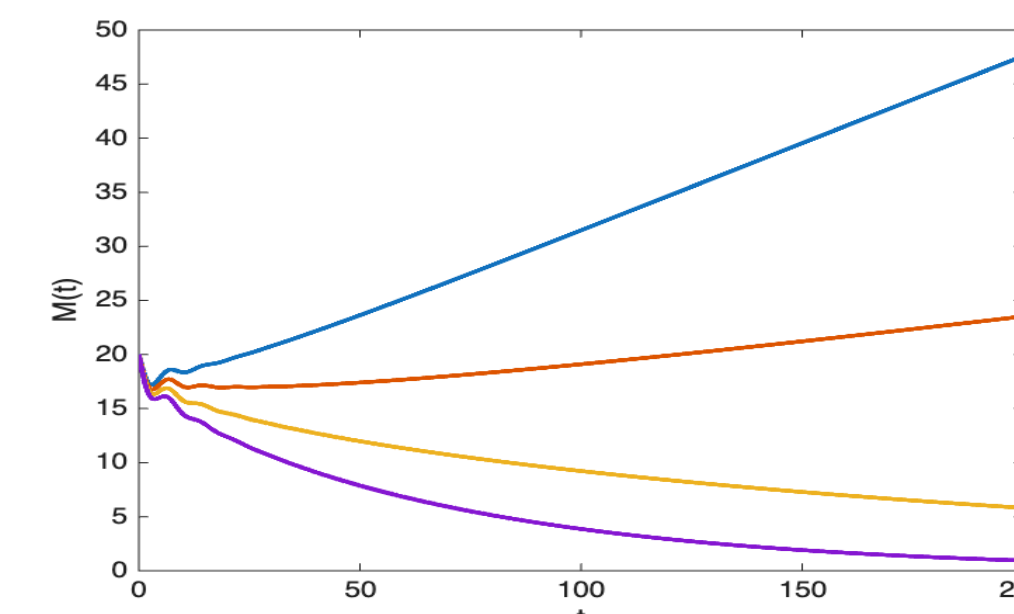


The asymptotic profile $\phi_{\infty}(\theta)$ for different birth rate functions $\nu(\theta, p)$ showing numerical convergence.

Expansion vs. Regression

Proposition 0.3. Assume that $m > 1$, that $\mu(\theta)$ satisfies $\int_0^{\infty} \mu(\theta) d\theta = \infty$ (every cell dies eventually) and that $\nu(\theta, p)$ satisfies $\partial_p \nu(\theta, p) < 0$, and $\nu(\theta, p_M) = 0$. Then the steady-state system of the model has a unique positive solution $\phi_{\infty}(\theta)$ if and only if

$$(0.2) \quad \int_0^{\infty} \mu(\theta) e^{-\int_0^{\theta} \nu(s, 0) + \mu(s) ds} d\theta < \frac{1}{2}.$$



Evolution of the mass of tumor for different values of the death rate μ both above and below the threshold $\mu_c = 0.0928$

- $\mu < \mu_c \rightarrow$ tumor expands
- $\mu > \mu_c \rightarrow$ tumor regresses

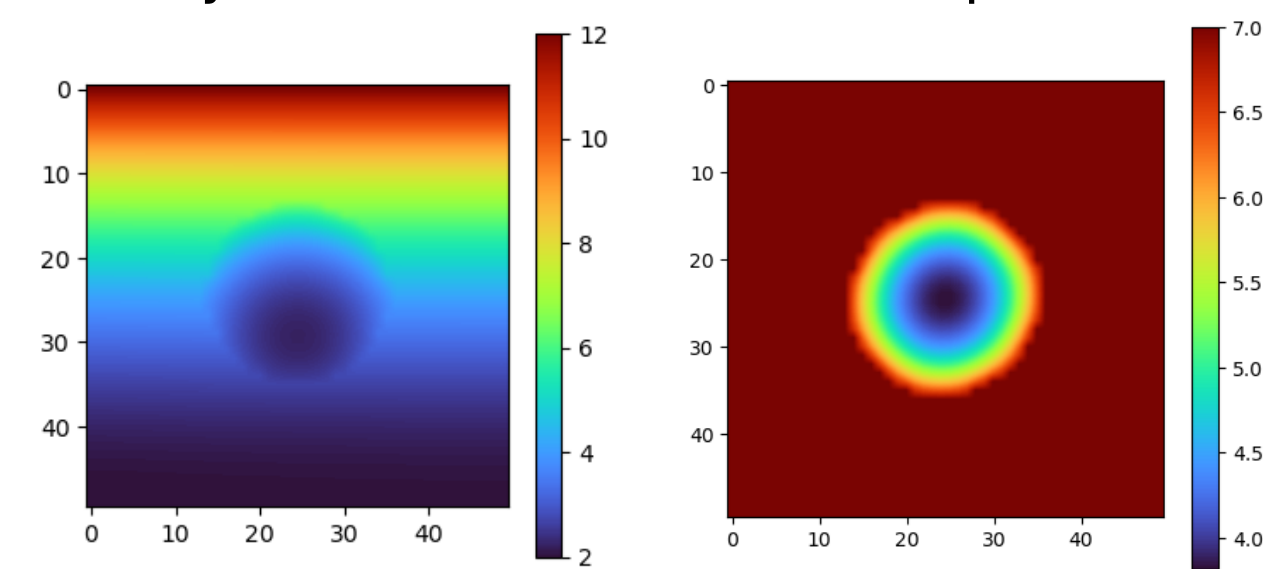
Extension to Evolution of Drug Resistance [2]

Hyunah Lim and Doron Levy

Model Description

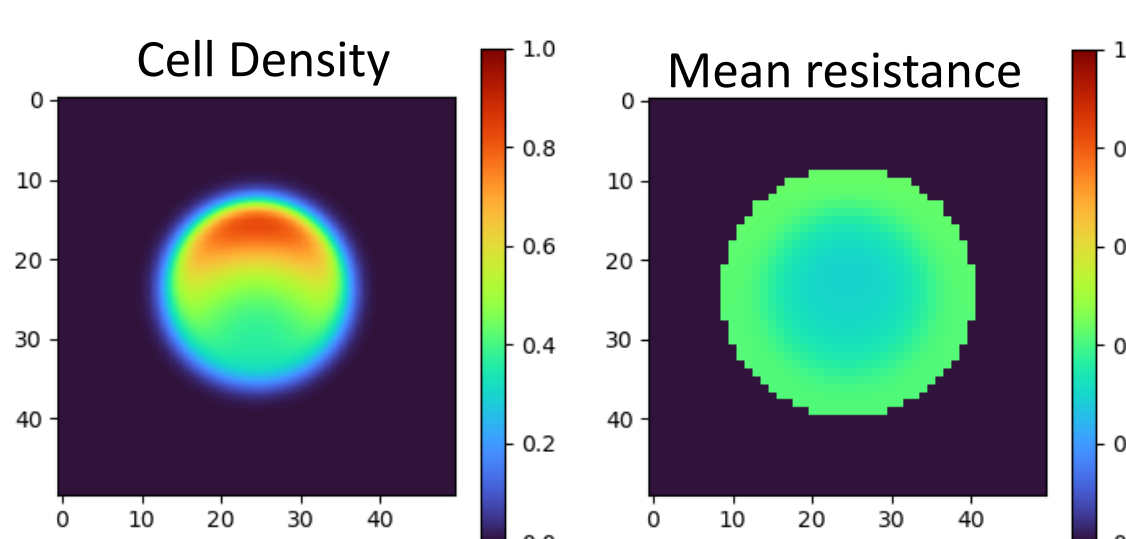
- Cell proliferation depends on cell age, local nutrient availability, and nutrient uptake rate
- Resistant cells pay fitness cost by lowered nutrient uptake rate
- Density saturation inhibits proliferation
- Drug-induced cell death depends on cell resistance phenotypes and local drug concentration
- Nutrient and drug distribution are modeled as a steady-state reaction-diffusion equation

Example nutrient and drug distributions under the boundary conditions: diffusion from the upper boundary (Left) and constant concentration outside tumor (Right)

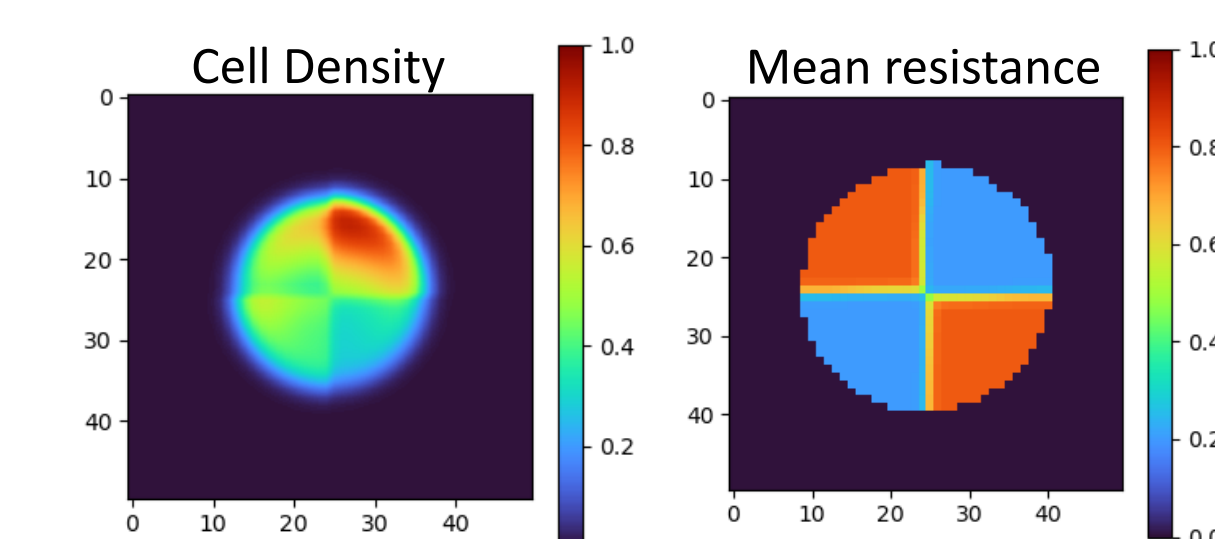


- Pretreatment tumor states generated from two resistance initial configurations:

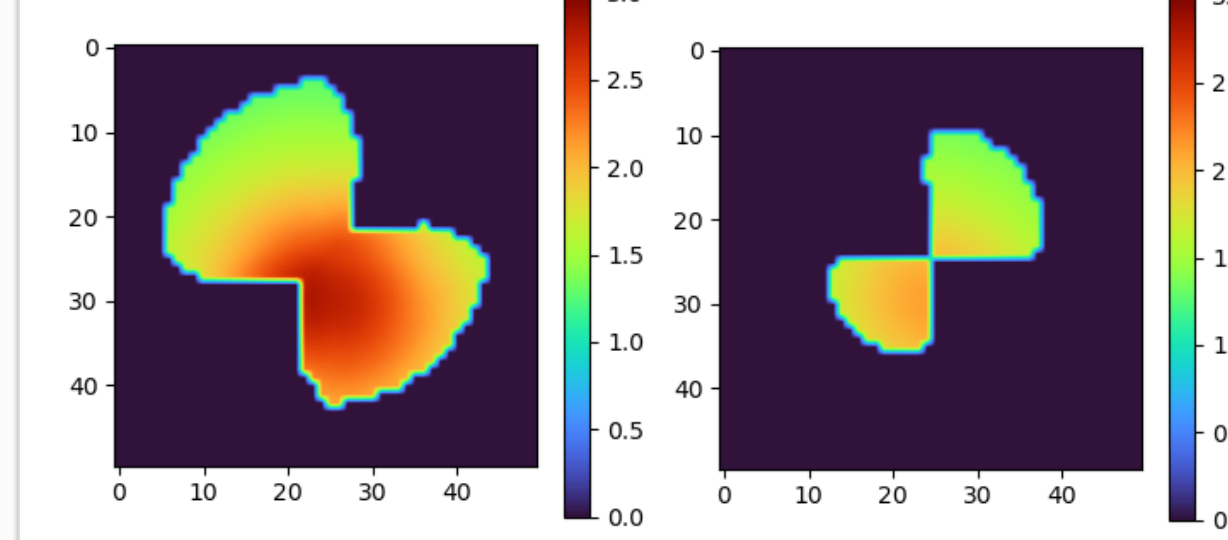
Spatially mixed resistance phenotypes



Spatially segregated resistance phenotypes



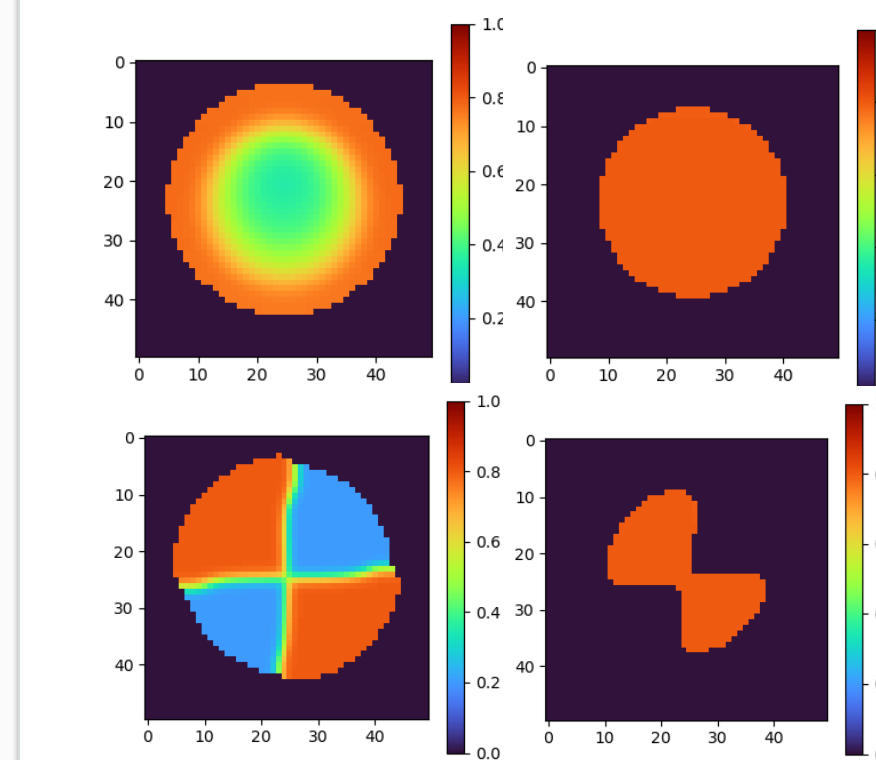
Average age



Average age of resistant (left) and sensitive (right) cells under $C_1^b = 4$, with nutrient and drug supplied by diffusion from the upper boundary and initially spatially segregated resistance phenotypes.

- Cells are younger on average in regions where nutrient and drug exposure are both higher, reflecting increased proliferation and treatment-induced death. Resistant cells remain older on average than sensitive cells because their lower drug-induced mortality allows longer persistence.

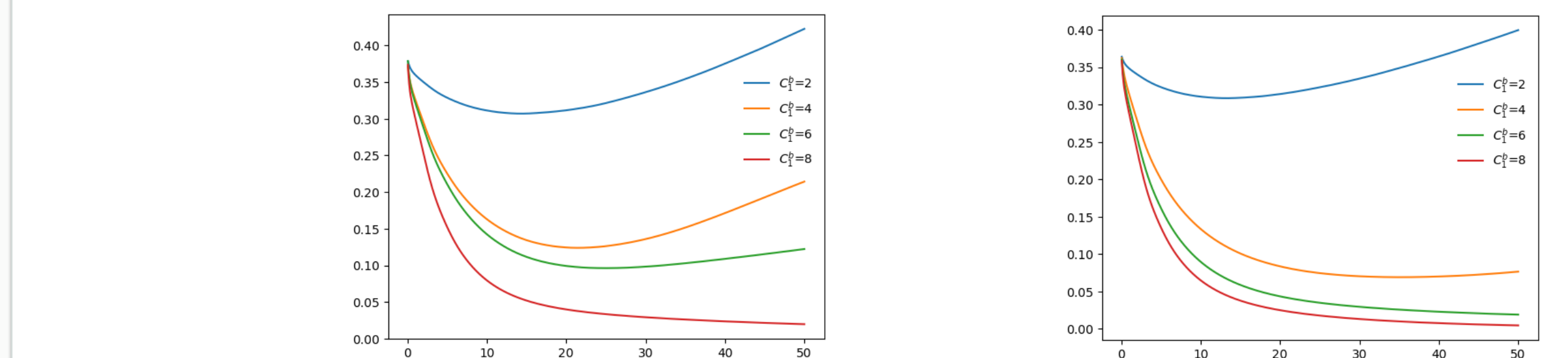
Resistance selection



Mean resistance under continuous chemotherapy. Upper row: tumors grown from spatially mixed resistance phenotypes; lower row: tumors grown from spatially segregated resistance phenotypes. Left column: $(C_1^b = 2)$; right column: $(C_1^b = 8)$. In all cases, nutrient and drug are supplied by diffusion from the upper boundary.

- The simulations show stronger selection for resistant phenotypes at the higher drug dose.

Total tumor burden



Total tumor burden under varying chemotherapy doses for spatially mixed and spatially segregated resistance phenotypes. In both simulations, nutrient and drug enter by diffusion from the upper boundary. **Left:** resistant and sensitive phenotypes are spatially mixed within the initial tumor. **Right:** resistant and sensitive phenotypes are spatially segregated within the initial tumor.

- Spatial coexistence improves tumor survival under treatment because the more drug-sensitive cells locally take up more drug, reducing drug availability to nearby resistant cells and thereby making the same administered dose less effective against the resistant population.

Conclusion

- Mechanical pressure shapes spatial patterns in cell age and proliferation in the model.
- The age distribution of our model numerically converges toward a stable long-time profile under our model.
- Finite speed of propagation of the tumor front is established analytically and numerically.
- The model predicts a threshold in the proliferation-death balance separating expansion from regression.
- In the drug-resistance extension, increased dose of chemotherapy reduces tumor burden while selecting for resistant phenotypes among surviving cells.
- Spatial organization of resistance can alter treatment efficacy

References

[1] Levy D, Lim H, Mellet A, Wildes M. *Age-Structured Mechanical Models for Tumor Growth*. Journal of Mathematical Biology (2026).

[2] H. Lim, *Age-Structured Models of Tumor Growth and Evolution of Drug Resistance*. Ph.D. dissertation, University of Maryland, (2025).

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